

COMPUTER NETWORKS

DATA LINK LAYER AND TRANSPORT BASICS

Descriptive questions, comparisons, examples, and revision notes

Exam Focus

How to use these notes. Each major item from the supplied topic file has been converted into an exam-style question and a descriptive answer. The conversational fragments, image placeholders, and file-reference markers have been removed, but the definitions, comparisons, examples, mechanisms, and exam conclusions have been retained and organized for study.

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1. TCP and UDP

Question 1: Compare TCP and UDP.

[10 marks]

Transmission Control Protocol (TCP) and **User Datagram Protocol (UDP)** are transport-layer protocols. They differ mainly in connection setup, reliability, ordering, control mechanisms, overhead, and typical applications.

Feature	TCP	UDP
Full name	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented	Connectionless
Setup	Establishes a connection before sending data	Sends data directly without connection setup
Reliability	Reliable delivery	No guarantee of delivery
Ordering	Ensures that data arrives in order	Packets may arrive out of order
Control	Error control, flow control, and congestion control	No built-in flow or congestion control
Speed	Slower because of extra checking and control	Faster because of less overhead
Recovery	Uses acknowledgements and retransmission	No acknowledgements or retransmission
Data unit	Segment	Datagram
Applications	HTTP/HTTPS, FTP, SSH, and email	DNS, video calls, online gaming, and streaming

TCP is selected when every byte must arrive correctly and in the correct order. UDP is selected when low delay is more important than retransmitting old data. For example, downloading a file uses TCP because every byte must arrive correctly. An online game or video call commonly uses UDP because speed matters more than recovering an outdated packet.

Remember

TCP prioritizes reliable and ordered delivery. UDP prioritizes speed and low overhead.

2. Bandwidth and Throughput

Question 2: Distinguish bandwidth from throughput.

[5 marks]

Bandwidth is the maximum possible data-carrying capacity of a network connection. It is a theoretical limit and is usually measured in bits per second (bps), megabits per second (Mbps), or gigabits per second (Gbps).

Throughput is the actual amount of data transferred successfully over the network. It is real-world performance and is affected by congestion, delay, errors, interference, protocol overhead, and other network conditions.

Feature	Bandwidth	Throughput
Meaning	Maximum possible capacity	Actual amount transferred
Nature	Theoretical limit	Real-world result
Measurement	bps, Mbps, Gbps	bps, Mbps, Gbps
Network conditions	Does not itself account for problems	Reduced by congestion, delay, errors, and interference

If an Internet connection has a bandwidth of 100 Mbps but a throughput of 70 Mbps, the connection can theoretically support up to 100 Mbps, while the actual received data rate is 70 Mbps because of network conditions.

Remember

Bandwidth = capacity. Throughput = actual speed achieved.

3. Guided and Unguided Transmission Media

Question 3: Explain guided and unguided transmission media. [5 marks]

Transmission media is the path through which data travels from a sender to a receiver.

3.1 Guided Media – Wired

In guided media, data travels through a physical cable. The main examples are twisted-pair cable, coaxial cable, and optical-fiber cable.

Type	Main feature
Twisted pair	Cheap and commonly used in Ethernet and LANs
Coaxial cable	Better shielding than twisted pair
Optical fiber	Uses light; provides very high speed and long distance

3.2 Unguided Media – Wireless

In unguided media, data travels through air or space using electromagnetic waves. Examples include radio waves, microwaves, infrared, satellite communication, Wi-Fi, and Bluetooth.

Type	Main feature
Radio waves	Used in Wi-Fi, radio, and mobile communication
Microwaves	Used in cellular and point-to-point links
Infrared	Used for short-range communication
Satellite	Used for long-distance or global communication

Feature	Guided	Unguided
Path	Physical cable	Air or space
Security	Usually more secure	More exposed to interference or interception
Deployment	Installation can be difficult	Easier to deploy in many places
Examples	Fiber and coaxial	Wi-Fi and satellite

Exam conclusion: Guided media transmits data through a physical medium, while unguided media transmits data wirelessly through electromagnetic waves.

4. Node-to-Node Communication and the DLL PDU

Question 4: Explain node-to-node communication in the Data Link Layer. [5 marks]

Node-to-node communication means transferring data from one directly connected device, or node, to another device over the same physical link. The Data Link Layer handles communication between neighboring nodes such as Computer to Switch, Switch to Router, and Router to Router.

The Data Link Layer performs the following tasks:

1. It frames the data into identifiable frames.
2. It adds source and destination MAC addresses.
3. It detects transmission errors using methods such as CRC.
4. It controls access to a shared medium.
5. It delivers the frame to the next directly connected node.

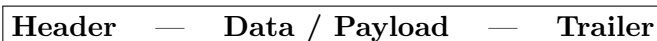
Computer → Switch → Router
Each Data Link Layer hop is a local, node-to-node transfer.

Do Not Confuse

Do not confuse the layers: the Data Link Layer provides node-to-node delivery, the Network Layer provides host-to-host delivery across networks, and the Transport Layer provides process-to-process delivery.

Question 5: What is the PDU of the Data Link Layer? Explain its structure. [5 marks]

The Protocol Data Unit (PDU) of the Data Link Layer is called a **frame**. A frame normally contains a header, a data or payload field, and a trailer.



- **Header:** contains source and destination MAC addresses and control information.
- **Data/Payload:** contains the packet received from the Network Layer.
- **Trailer:** usually contains error-detection information such as FCS or CRC.

During encapsulation, a Network Layer packet is placed inside the payload of a Data Link Layer frame. The exam line is: *A frame is the PDU of the Data Link Layer.*

5. Services Offered by the Data Link Layer

Question 6: Explain the services offered by the Data Link Layer. [10 marks]

The Data Link Layer provides services for transferring frames between two directly connected nodes. Its services include three delivery services and the functions of framing, flow control, error control, and congestion control.

5.1 Unacknowledged Connectionless Service

Frames are sent independently. No logical connection is established before transmission, and the receiver does not send acknowledgements. If a frame is lost or damaged, the Data Link Layer does not retransmit it. Basic Ethernet service is an example.

5.2 Acknowledged Connectionless Service

No logical connection is established, and each frame is sent independently. However, the receiver sends an acknowledgement for each correctly received frame. If a frame is lost, the sender may retransmit it. This service is suitable when reliable delivery is required but establishing a connection is unnecessary.

5.3 Acknowledged Connection-Oriented Service

A logical connection is established between the sender and receiver before data transfer. Frames are usually numbered, acknowledged, and delivered in the correct order. Lost or damaged frames can be retransmitted. After transmission, the connection is terminated. This service provides more reliable and organized frame delivery.

5.4 Framing

The Network Layer packet is divided into smaller units called frames. Frames contain control information such as source MAC address, destination MAC address, and error-detection information. Framing helps the receiver identify the beginning and end of each frame.

5.5 Flow Control

Flow control controls the rate at which data is transmitted. It prevents a fast sender from overwhelming a slow receiver and ensures that the receiver has enough time and buffer space to process incoming frames.

5.6 Error Control

Error control detects corrupted, lost, or duplicated frames. It uses techniques such as CRC, checksum, acknowledgements, retransmission, and timers. These mechanisms help maintain accurate data transfer between nodes.

5.7 Congestion Control

Congestion control manages excessive traffic on a link or network. It prevents too many frames from being transmitted at the same time and helps reduce frame loss, transmission delay, and network overload.

Service	Logical connection	Acknowledgement	Reliability
Unacknowledged connectionless	No	No	Low
Acknowledged connectionless	No	Yes, per frame	Moderate/High
Acknowledged connection-oriented	Yes	Yes	High

Exam conclusion: The Data Link Layer offers unacknowledged connectionless, acknowledged connectionless, and acknowledged connection-oriented services. It also performs framing, flow control, error control, and congestion control to support efficient and reliable node-to-node communication.

6. Point-to-Point and Broadcast Links

Question 7: Compare point-to-point and broadcast links.

[5 marks]

A **point-to-point link** connects exactly two devices. Only two nodes share the link, data is sent directly from one device to the other, and no other device competes for the link. Router-to-router connections and leased lines are common examples. A direct cable connecting two routers is a point-to-point link.

A **broadcast link** connects multiple devices to the same shared communication medium. A frame transmitted by one device may be received by all connected devices. The destination device processes the frame while other devices ignore it. Such a link requires access control to decide which device can transmit. Traditional shared Ethernet and Wi-Fi are examples.

Feature	Point-to-point	Broadcast
Devices	Connects two devices	Connects multiple devices
Medium	Dedicated link	Shared medium
Competition	No competition between many devices	Devices may compete for access
Example	Router-to-router cable	Wi-Fi network

Exam conclusion: A point-to-point link provides a direct connection between two nodes, whereas a broadcast link allows multiple nodes to share the same communication medium.

7. Data Link Layer Sublayers

Question 8: Explain the LLC and MAC sublayers.

[10 marks]

The Data Link Layer is divided into two sublayers because it has a double job: it communicates with the software or Network Layer above it, and it handles the physical transmission medium below it.

7.1 LLC – Logical Link Control

The LLC sublayer is the software and logic side of the Data Link Layer. It communicates with the Network Layer and provides logical control over the data link.

Its functions are:

- Identifies and manages Network Layer protocols.
- Provides flow control.
- Provides error control in supported protocols.
- Handles communication between the Network Layer and the MAC sublayer.

For example, LLC can identify whether the carried packet belongs to IPv4 or IPv6.

7.2 MAC – Media Access Control

The MAC sublayer is the hardware and physical-medium side of the Data Link Layer. It controls how devices access the transmission medium.

Its functions are:

- Adds source and destination MAC addresses.
- Controls access to a shared medium.
- Creates and processes frames.
- Helps detect transmission errors using FCS or CRC.
- Handles collisions or channel access depending on the technology.

Examples include Ethernet MAC and Wi-Fi MAC.

Feature	LLC	MAC
Full name	Logical Link Control	Media Access Control
Main relationship	Communicates with the Network Layer	Communicates with the Physical Layer
Main function	Logical link management and upper-protocol identification	Medium access, framing, and MAC addressing
Typical association	IEEE 802.2	Ethernet and Wi-Fi MAC
Memory question	Which network protocol is being carried?	Who can transmit, and where should the frame go?

8. Link-Layer Addressing

Question 9: Explain link-layer addressing and distinguish IP and MAC addresses. [10 marks]

The Data Link Layer uses MAC addresses to deliver a frame across the current local link. An IP address identifies the final source and destination hosts across an end-to-end path, whereas a MAC address identifies devices involved in the current local hop.

Address	Used for	Scope
IP address	Identifies final source and destination hosts	End-to-end
MAC address	Identifies devices on the current link	Local hop

Suppose Computer A sends data to Server B through two routers. The IP source and destination remain associated with the end-to-end communication, but the MAC source and destination change at each hop:

Hop	Source MAC	Destination MAC
Computer A to Router 1	Computer A	Router 1
Router 1 to Router 2	Router 1	Router 2
Router 2 to Server B	Router 2	Server B

A new frame is created on every hop. Therefore, IP addresses identify the end-to-end communication, while MAC addresses identify the devices involved in the current local link.

Do Not Confuse

The IP destination does not change during routing, but the MAC source and destination can change at every router-to-router or host-to-router link.

9. NAT, PAT, and Port Translation

Question 10: Explain how NAT/PAT changes source addressing while accessing HTTPS. [10 marks]

When a device connects to a secure website, the destination port is normally 443. The destination port identifies the service on the server and remains 443 as the packet leaves the home network. The browser usually chooses a random high-numbered source port, such as 54321, to receive the reply.

When the packet reaches the home router, NAT, specifically Port Address Translation (PAT), changes the source IP address and source port. This is necessary when multiple devices inside the home network access the same Internet service at the same time. They may all use destination port 443, so the router creates unique public-side source ports to identify each conversation.

9.1 NAT Table Example

Field	Step A: inside the house	Step B: after the router
Source IP	192.168.1.50	99.100.101.102
Source port	54321	9001
Destination IP	8.8.8.8	8.8.8.8
Destination port	443	443

The router records a mapping such as: anything returning from the server to public port 9001 belongs to the internal device at 192.168.1.50:54321. The router uses this table to send the reply to the correct internal device.

Remember

For this HTTPS example, the destination port stays 443. PAT changes the source port to a unique tracking port and also replaces the private source IP with the router's public IP.

10. Unicast, Multicast, and Broadcast

Question 11: Explain unicast, multicast, and broadcast delivery. [5 marks]

These terms describe how data is delivered from a sender to receivers.

10.1 Unicast – One-to-One

Unicast sends data from one sender to one specific receiver. Examples include sending a message to one person, loading a website from a server, and a computer sending data to a specific router. At the MAC level, the pattern is one source MAC address to one destination MAC address.

10.2 Multicast – One-to-Many Selected Group

Multicast sends data from one sender to a specific group of receivers. Only devices that joined the multicast group receive or process the data. Examples include online live streaming to subscribed users, video conferencing, and routing protocol updates.

10.3 Broadcast – One-to-All on the Local Network

Broadcast sends data from one sender to all devices on the local network. All devices receive the frame, but only the relevant devices may process it. Examples include an ARP request in IPv4 Ethernet and a DHCP Discover message. The Ethernet broadcast MAC address is `FF:FF:FF:FF:FF:FF`.

Type	Delivery	Example
Unicast	One to one	Website request
Multicast	One to selected group	Live streaming group
Broadcast	One to all local devices	ARP request

Do Not Confuse

A point-to-point link describes the type of connection between two nodes. A broadcast link describes a shared medium used by multiple nodes. Unicast, multicast, and broadcast describe methods of delivering data.

11. Framing and Framing Methods

Question 12: What is framing and why is it needed? [5 marks]

Framing is the process of dividing a continuous stream of bits received from the Physical Layer into smaller, identifiable units called frames.



The Data Link Layer needs to identify where a frame starts, where it ends, which bits belong to each frame, and whether the frame contains errors. The DLL divides the bit stream into discrete frames and may calculate a checksum for each frame to detect errors.

The main framing methods are byte count, flag bytes with byte stuffing, and flag bits with bit stuffing.

Question 13: Explain the byte count method. [5 marks]

In the byte count method, the frame header contains a field indicating the number of bytes in the frame. If the count field says 5, the receiver reads the count and knows that the frame contains five bytes of data.



The working is:

1. The sender places the frame length in the count field.
2. The receiver reads the count.
3. The receiver collects that number of bytes.
4. The frame ends after the specified number of bytes.

The disadvantage is that if the count field becomes corrupted, the receiver may lose frame synchronization. For example, if the actual count is 5 but the corrupted count is 8, the receiver may read bytes belonging to the next frame as part of the current frame.

Question 14: Explain flag bytes with byte stuffing. [10 marks]

This method uses a special flag byte to mark the beginning and end of a frame.



For example, a frame may be represented as `FLAG A B C D E FLAG`. The receiver knows that the frame starts and ends at the flag byte.

The problem occurs when the flag byte appears inside the actual data. In **FLAG A B FLAG C D FLAG**, the receiver may mistakenly think that the middle flag marks the end of the frame.

The solution is **byte stuffing**. The sender inserts an escape byte before any flag byte appearing in the data. Suppose **F** is the flag byte and **E** is the escape byte:

Original data: A B F C D
 After byte stuffing: A B E F C D
 Complete frame: F A B E F C D F

When the receiver sees **E F**, it understands that this is an actual data byte **F**, not the end-of-frame flag. The receiver removes the escape byte; this process is called byte de-stuffing.

- The flag byte marks frame boundaries.
- The escape byte protects flag-like data.
- The sender inserts escape bytes.
- The receiver removes escape bytes.

Question 15: Explain flag bits with bit stuffing.

[10 marks]

This method uses a special bit pattern as the beginning and ending flag. A common flag pattern is 01111110.

01111110 DATA 01111110
 Flag Flag

The problem is that the flag pattern may accidentally appear inside the data. To prevent this, the sender uses bit stuffing.

Bit stuffing rule: Whenever the sender finds five consecutive 1s in the data, it automatically inserts a 0 bit. The receiver removes that inserted zero during de-stuffing.

For example, if the data contains a sequence that includes five consecutive 1s, the sender inserts a zero after those five 1s. Conceptually:

Original: 111110
 Stuffed: 1111100

When the receiver detects the stuffed zero after five consecutive 1s, it removes the zero. This process is called bit de-stuffing. Bit stuffing works at the bit level, prevents data from being confused with the flag pattern, and maintains frame synchronization.

11.1 Byte Stuffing versus Bit Stuffing

Feature	Byte stuffing	Bit stuffing
Works on	Bytes or characters	Individual bits
Uses	Flag byte and escape byte	Flag bit pattern
Stuffing rule	Insert escape before flag-like byte	Insert 0 after five consecutive 1s
Removal	Byte de-stuffing	Bit de-stuffing
Example	E F represents data F	A stuffed zero is removed after five 1s

Exam conclusion: Framing divides a continuous bit stream into identifiable frames. The major framing methods are byte count, flag bytes with byte stuffing, and flag bits with bit stuffing. Byte count uses a length field, byte stuffing uses escape bytes to protect flag bytes, and bit stuffing inserts a zero after five consecutive ones to prevent confusion with the flag pattern.

12. Important Exam Topics

- TCP versus UDP: connection, reliability, ordering, control, overhead, and applications.
- Bandwidth versus throughput, including the 100 Mbps versus 70 Mbps example.

- Guided and unguided media with examples and differences.
- Node-to-node communication and the Data Link Layer frame PDU.
- The three DLL delivery services and the functions of framing, flow control, error control, and congestion control.
- Point-to-point versus broadcast links.
- LLC versus MAC sublayers.
- IP versus MAC addresses across multiple hops.
- NAT/PAT source-port translation for HTTPS traffic.
- Unicast, multicast, and broadcast.
- Framing, byte count, byte stuffing, bit stuffing, and de-stuffing.

13. 10-Mark Answer Templates

Question A: Explain the Data Link Layer and its services.

[10 marks]

Answer Skeleton

Write in this order:

1. Define node-to-node communication and state that the DLL PDU is a frame.
2. Draw or describe the frame structure: header, payload, and trailer.
3. Explain unacknowledged connectionless service.
4. Explain acknowledged connectionless service.
5. Explain acknowledged connection-oriented service.
6. Explain framing, flow control, error control, and congestion control.
7. Add the three-service comparison table.
8. Conclude that the DLL supports efficient and reliable frame transfer between directly connected nodes.

Question B: Explain the LLC and MAC sublayers.

[10 marks]

Answer Skeleton

Define why the DLL is divided into two sublayers. Explain LLC as the logical/software side that communicates with the Network Layer, identifies upper protocols, and supports control. Explain MAC as the medium/hardware side that handles MAC addresses, frames, access to the medium, collisions, and FCS/CRC. Add a comparison table and finish with the memory questions: “Which network protocol is being carried?” for LLC and “Who can transmit, and where should the frame go?” for MAC.

Question C: Explain framing methods.

[10 marks]

Answer Skeleton

Define framing and explain why start, end, membership, and error detection are needed. Explain byte count with its count field and synchronization problem. Explain flag bytes with byte stuffing, including the escape-byte example. Explain flag bits with bit stuffing, including the five-1s rule and de-stuffing. End with a byte-stuffing versus bit-stuffing comparison.

Question D: Explain link-layer addressing and NAT/PAT.

[10 marks]

Answer Skeleton

Distinguish IP scope from MAC scope. Use the Computer A to Router 1 to Router 2 to Server B example to show that a new frame is created at every hop. Then explain the HTTPS example: destination port

443 remains the service port, while PAT changes the private source IP and source port to a public address and unique tracking port. Include the NAT table mapping and explain how the return traffic is delivered to the correct internal host.

14. Question Bank

Definitions / 2–3 marks

1. Define TCP and UDP.
2. Define bandwidth and throughput.
3. What is transmission media?
4. Define node-to-node communication.
5. What is the PDU of the Data Link Layer?
6. Define framing.
7. What is a point-to-point link?
8. What is a broadcast link?
9. Expand LLC and MAC.
10. What is a MAC address?
11. Define unicast, multicast, and broadcast.
12. What is byte stuffing?
13. What is bit stuffing?

5-mark questions

1. Compare TCP and UDP.
2. Distinguish bandwidth from throughput using a numerical example.
3. Explain guided and unguided transmission media.
4. Explain the structure of a Data Link Layer frame.
5. Explain the three Data Link Layer delivery services.
6. Compare point-to-point and broadcast links.
7. Explain LLC and MAC sublayers.
8. Compare unicast, multicast, and broadcast.
9. Explain the byte count framing method.

10-mark questions

1. Explain the services offered by the Data Link Layer.
2. Explain the LLC and MAC sublayers with functions and comparison.
3. Explain framing methods: byte count, byte stuffing, and bit stuffing.
4. Explain link-layer addressing across multiple router hops.
5. Explain NAT/PAT using the HTTPS destination-port and source-port example.
6. Compare TCP and UDP and discuss their applications.
7. Explain node-to-node communication, frame structure, and Data Link Layer responsibilities.

15. Last-Minute Revision Page

TRANSPORT

TCP is connection-oriented, reliable, ordered, and controlled. UDP is connectionless, fast, and low-overhead.

CAPACITY

Bandwidth is maximum capacity. Throughput is actual achieved transfer.

MEDIA

Guided uses cable. Unguided uses electromagnetic waves through air or space.

DLL PDU

Frame = header + payload + trailer. Header has MAC addresses; trailer commonly has FCS/CRC.

DLL SERVICES

Unacknowledged connectionless, acknowledged connectionless, acknowledged connection-oriented; plus framing, flow, error, and congestion control.

SUBLAYERS

LLC talks to the Network Layer. MAC controls the medium, addresses, and frames.

ADDRESSING

IP is end-to-end. MAC is local-hop. A new frame is created at every hop.

NAT/PAT

HTTPS destination port 443 stays 443; the router

changes private source details to public source details and records a mapping.

DELIVERY

Unicast = one to one; multicast = one to selected group; broadcast = one to all local devices.

FRAMING

Byte count uses length. Byte stuffing uses an escape byte. Bit stuffing inserts 0 after five consecutive 1s.

EXAM DIAGRAMS

Frame structure, hop-by-hop MAC addresses, NAT table, flag-byte frame, and flag-bit frame.

Source Scope

This document is based on the supplied file `/home/elish4h/Rajagiri/CNIA/todo-notes-topics.txt` and follows the structure of `/home/elish4h/Rajagiri/notes-template.tex`. The source file contained conversational fragments and image/content-reference placeholders; these have been removed or converted into ordinary explanatory text while preserving the substantive definitions, examples, comparisons, working steps, and exam conclusions. No placeholder answer has been left for a topic present in the supplied notes.

Study principle: understand the mechanism first, then memorize the exam wording.